

Develop A Fastest Finger First System For Six Players

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Described here is an Arduino-based fastest finger first system. In quiz shows, the buzzer or fastest finger first round is frequently used to determine which candidate (or group) achieves the highest priority in attempting the question. The fastest finger first method involves candidates pressing a key (or buzzer) placed in front of them such that whoever presses the key first gets the right to answer the question first.

Here, the buzzer is emulated by using a keyboard having six switches indicating six candidates. Arduino continuously scans the keys and records the key numbers, one through six, to see which one is pressed first. Key numbers are also displayed on an LCD to indicate which key is pressed first. The authors' prototype of the project is shown in Fig. 1.

Circuit and working

The circuit diagram of the fastest finger first system using Arduino is shown in Fig. 2. It is built around Arduino Uno board (board1), 16x2 alphanumeric LCD (LCD1), six tactile switches and a few other components.

Main components used in this project are:

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and fourteen digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM.

Keyboard. In this project, a 1x6 matrix keyboard is used as the input device. The row is connected to ground and the six columns are connected to Arduino analogue input pins (A0 through A5) and 5V individually through 1-kilo-ohm resistor.

LCD. A 16x2 LCD is connected to digital pins 12, 11, 5, 4, 3 and 2 of Arduino to interface with it. The first

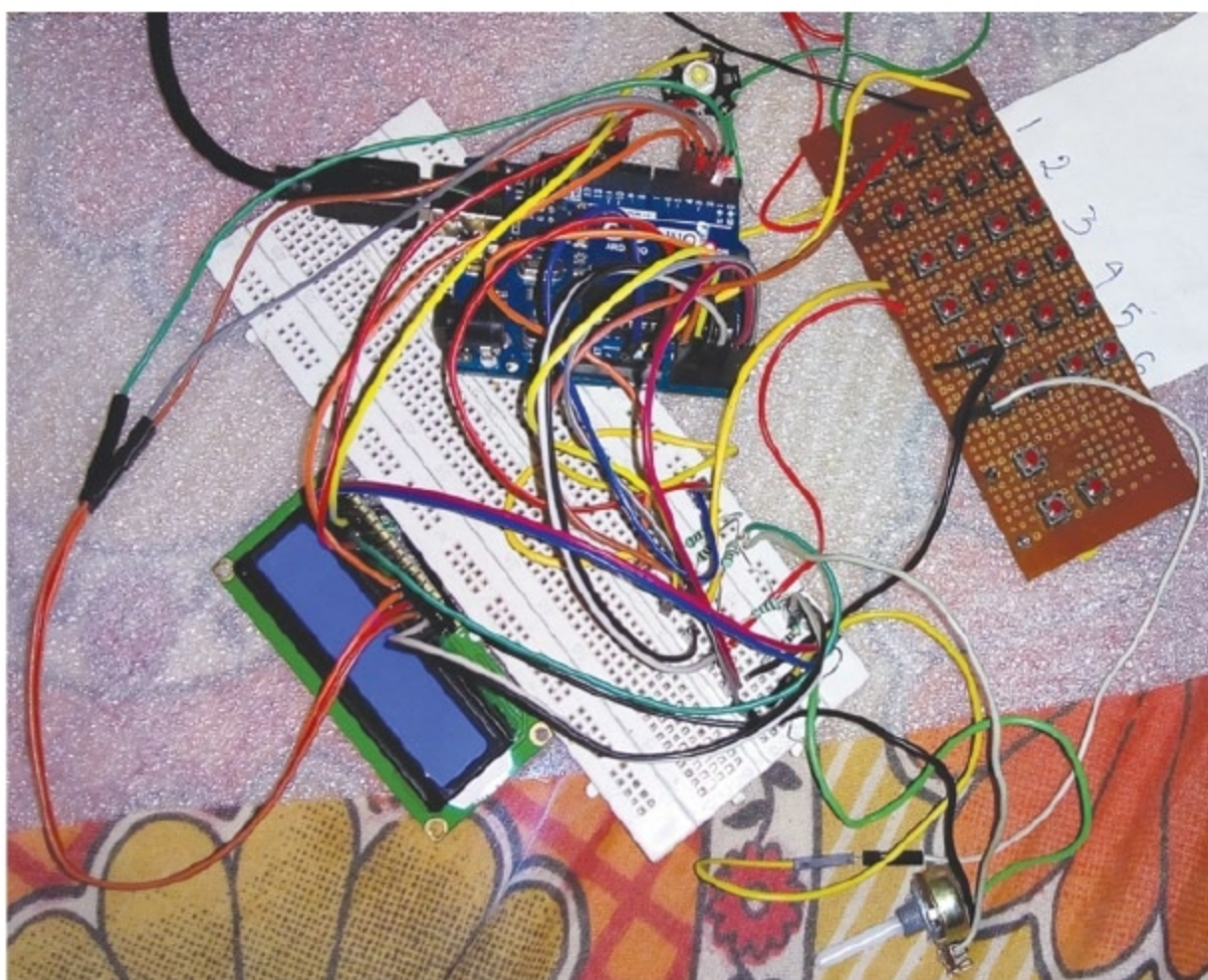


Fig. 1 : Authors' prototype of the fastest finger first system using Arduino

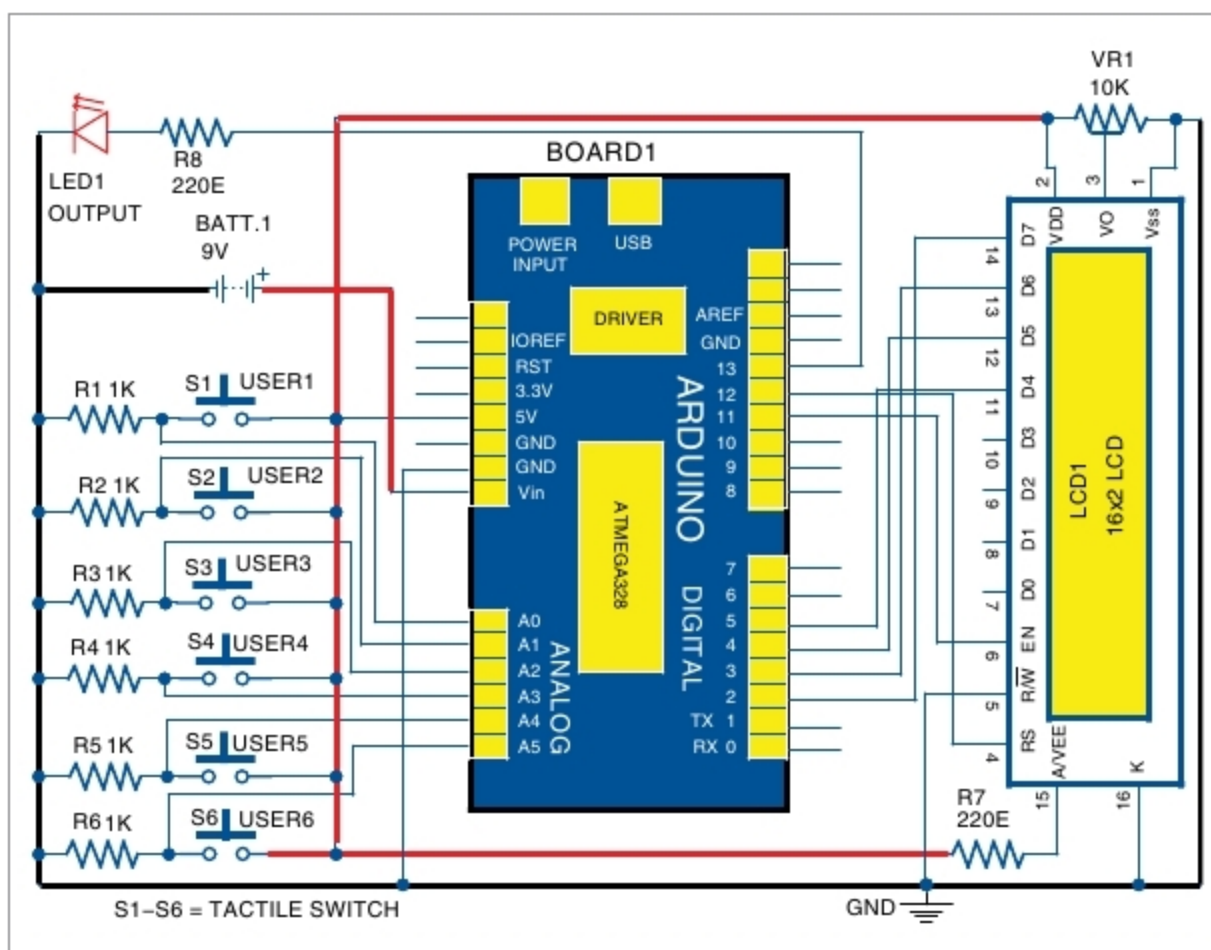


Fig. 2: Circuit diagram of the project